ActInf GuestStream #007.1 ~ How particular is the physics of the Free Energy Principle?

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[Music] hello and welcome everyone to the actin flab guest stream number 7.1.07.1 actually very nice number it's may 28 2021 and we have some very special guests on today who i will let themselves uh introduce themselves and then we'll go into a presentation followed by a discussion so any and all questions and comments just put into the live chat during the presentation and i'll be compiling those so that we can make sure to ask them so please miguel go for it okay hi yeah i'm miguel aguilera i'm working as a postdoc since a year now at the university university of sussex and then so my name is christopher buckley i'm a lecturer at the university of sussex in both the sussex neuroscience and sussex ai groups uh interested in sensory motor control in general cool so miguel um share your screen i'll resize it and then we'll have a presentation take as long as you'd like okay okay thanks a lot for the yeah for the invitation i'm really happy to present our work here and it's been really kind of fast because we just released the prayer print a few days ago but it's and it's good for the opportunity to do this discussion online so far uh okay let's see okay can you see my screen yep so and i'll resize it now so go for it okay okay so okay so this is this work uh so this is a problem that we have just published as miguel aguilera and and i we wrote it i wrote it with uh alexander alexander sons and christopher buckley from all from sussex but what's except for brenda's now in oslo um so the um so that they work just to focus on some of the recent publications about the free energy principle i guess most people watching know that the pre-inner principle is was conceived originally as a unified brain theory integrating experimental data and trying to listen to describe the relationship between action perception learning and so on and more recently the theory has evolved and and tried to become more general stating that any non-equilibrium dynamical system under some special conditions can be interpreted as performing by using inference and this will be a broader theory of living system or a way to describe how certain living systems behave using the tools from magician inference and and okay so so this is kind of so in principle this this kind of schema of how a brain is predicting an environment could be using different systems like a cell or different human beings and what i i imagine anyone familiar with with the primary principle have seen this kind of pictures a lot and we're trying to to to do different ones and yeah so i think it's a maybe a good thing to to to try to visualize things in a new ways because these ones have been repeated done so many times um okay so so the idea is so we are interested in in exploring this the the different assumptions behind the principle because um in in theory the free energy principle applies to a very diverse systems but in practice there has been very few attempts to explore it to explore all the steps in the of the theory in specific systems and what we want to do here is just that and we want to understand the free energy principle and its mathematical assumptions in the simplest system in the simplest non-equilibrium system that we can find and this is a linear language dynamics that i will explain later and under an assumption of and the assumptions that the couplings between elements are weak and in this very special case we can compute all the relevant variables analytically and this will be a very again this is a very interesting case for studying in in-depth all the mathematical details um okay so the the work i'm presenting today has is related to the latest publications around the ideas of the fib the monograph of free energy for a particular physics and the most recent more recent paper uh by thomas lance and carl about markov blankets information geometry and stochastic thermodynamics and specifically we want to focus on on two questions the third question is is how general is that is uh from all the systems that we can imagine can capture aspects of biological systems how many of them conform to the assumptions that that the the principle requires and the second question is how informative is the principle about the behavior of a system so if if a system if the first point holds so if a system conforms to all the requirements of the principle then how much how much do we know about the system right how much uh the principle is able to inform us about the behavior of this organism okay so so and and in this this we we find two issues that i will explain later in detail one is that and some of the requirements of of the principle uh are are make it challenging to find systems in in the class of systems we explore that that find that that match the required assumptions and and the second issue we found is that one important step in connecting the behavior of the system to the to the gradient of the free energy of the sherlock principle uh involves uh the coupling from the history of interaction of a system and this creates problems in the for interpretation of a system behaving as if minimizing apprentice i will explain that into that so so before starting just one quick note on the notation we are using which is a bit different from from the papers and some of the literature we are describing external sensory active and internal state with dc which is composed by external states y sensory states s active states a and interval states x and here also differently than in the in the papers both symbols represent a vector and matrices so and expectations instead which were revolved in some of the papers here are represented by these m variables like my ms and so on and sometimes the theory uses conditional expectations given a blanket state and in this case the the blanket states the conditional state is going to be between parentheses so this is the mean of y the mean of external states given b and also we can compute time derivatives as usual and we will represent marginal flows as these f variables that i will describe later okay so so first we try to summarize the different conditions and assumptions that one needs to derive the free energy and i am going to make a bit summary which we will be useful for later so first we the free energy principle assumes we have a system that we can as a langevin stochastic differential equation and this is just having a noise term this omega which is just random and uncorrelated the white noise and some flow function f that could be any potentially non-linear function that describes this time derivative this evolution this deterministic part of the evolution of of a variable c and in principle the the cubar er omega's gaussian term and its covariance is just a diagonal matrix a gamma because all the noise is independent between elements okay so the first condition that the the free energy principle requires is that the flow function decouples what the it's generally called autonomous and non-autonomous states meaning that internal states x are not going to influence external and sensory states and external states are not going to influence active and internal states and this so this is what uh what we generally expect from a sensory motor loop in an agent and so if we happen we represent this evolution of the system like an euler integration so the external states do not influence active and internal and internal states do not influence external and sensor okay so so we are going to require restrictor analysis to system that have this sensory model structure the second condition of the principle is that this system has a global attractor that is that the the probability distributions of the state of a system converge to some specific probability distribution which is called t and this speed and can be described by the surprise which is the negative logarithm of this of the probability which is how surprising or how unlikely is a state and and the con this condition proposes that this global attractor or this steady-state distribution can be described by a decomposition composition helpful holds the composition or well an a of the composition which relates the flow function with the surprise via two matrices q and gamma gamma we saw it before it was just the the covariance of the noise and q is a matrix capturing what's generally called the solenoidal flows in the system and q is a it's asymmetric and the solenoidal flows of the system mean mean that uh it capture the non-equilibrium influences in the system if if if a system is at thermodynamic equilibrium then q is going to be zero for all elements but if the system has an exchange of matter or energy or have some asymmetry in the interaction between the parts q is going going to be non-zero okay okay so we are going to have this and this diffusion turns gamma which are these these dot these dust lines and then we are going to have couplings between elements with these solenoidal influences that are captured by these uh blue arrows here this one will be arrows okay so that's the second condition and the last condition that the free energy sport requires is the mark of blanket condition this yeah maybe very well known which requires that so that the sensory motor states s and a and which are generally called a blanket state with that we are going to call v compressing the two so the condition requires that this variable and are going to play a very special role decoupling internal external states in the following way and they are going to if if we compute a conditional distribution fixing b and of into external internal states these states is going to be independent for a fixed v so they are not independent in general but they are independent for a specific blanket state okay so we can yeah so if these arrows are conditional our statistical dependencies the given b we have y and x but they are not and they are not statistically correlated for a given b okay and and so and in this sense the these blanket states are going to be like a a boundary or a barrier between external and internal states okay so so these are the conditions required by the free energy principle in orbit meaning that these these are the prerequisites that we require so for a system to uh to to to display the properties uh proposed by the free energy principle okay but asides of that uh we interpret and that the theory has some three extra assumptions so these are not conditions these are not presuppositive but things that the theory expects to happen or expect to find at least in some some cases and and for explaining this we still explained before and how the theory proposed that and variational free energy is involved in the behavior of a system so so the free energy principle proposes that any system under these conditions that they describe can be seen as minimizing or as if it's going to behave as if minimizing a variational free energy so for and for defining this free energy we start with the surprise of the blanket state so the blanket state is is what an agent can observe in the environment the agent uh do not cannot access the environment directly because of the essential metastructure that we have but only can access the blanket states the sensors and the actuators and and if a system is is behaving as if minimizing the free energy or and doing variation inference assistant will is going to behave as if minimizing the um yeah the the surprise of this this state and okay but the but the system cannot access this probability probability distribution directly but because it doesn't really cannot access the environment so instead of that by using inference prescribes to use a lower bound of this surprise and okay which is called a variational free energy and this variation of free energy consists of on the surprise term plus some divergence with some kl divergence between the actual probability distribution of the environment given the blanket and a model q a simplified model of the environment and and by ambassador inference consists in finding the parameters theta of this model that minimize this distance so in this so when this distance is minimized then the agent has a model of the world and is predicting the the states of the environment white okay so so solving the digestion inference problem uh so the simplest way of simplest way of doing it is just doing a gradient descent minimization of this free energy which ends up in being a gradient descent of of the surprise of the environment y at at the most likely state theta and blanket okay so so here just theta just parameterizes the most likely state of y okay um okay so so then the the frequency principle defines the most likely states of the uh of the system and environment peaks in a blanket just like the the mode or the arc max but sometimes it has been proposed that the mean is enough for doing this and since we are going to use linear systems they both are going to be the same so you can interpret this m as the mode or as the as the mean of of y and x given b and then you can if you remember we have these flow functions f you can compute the average flow of the system and y for a given b and which we saw in this assumption that can be described by this matrix q and gamma and the gradient on the surprise so if we want to find something like the gradient of the surprise that we have here and free energy principle proposes that we have to compute the average of of the flow given b and we obtain something like this and and we can see here that the first term is just what we want is the gradient is sorted with the gradient of the surprise but we have some extra terms um and so if we have so if we have some some gradient like we could if we had something like the first term in this equation we could interpret we can use this orient to perform um we will use this way to perform a minimization of this energy which interpret if we interpret it and strictly it could mean having some variable theta that follows this gradient oh sorry one second and okay and okay so so if we have some okay so and the effect the free energy so suggest that this parameters theta is going to be the most likely states of external or then internal i'm going to do so so we have sorry we have this equation here this equation 12 and we just need the first term so in general the free energy principle proposes that uh the the couplings of this matrix q the non di um are going to be zero and okay so so okay so this is uh so each element in this diagonal is a block can have many a number of dimensions and the free energy principle assumes well one interpretation is that it assumes that removing these solenoidal couplings these non-block diagonal copings then we are going to remove these elements that we don't want and we are going to have a flow of the system uh equal to this gradient on the surprise that is the gradient that minimize the free energy okay so so this is the first assumption if if this if these q terms are equal to zero then we have an average flow that goes in the direction of minimizing the free energy and this is interesting because we will have a system that points in the uh in the direction of the free energy and what we i will explain that later and however there's in some works uh it is proposed that the there's a more general form of this matrix that is it's also block diagonal but not within y and sna but within uh autonomous and non-autonomous and a and x and and this is has been referred as the general case but it's not for us at least it's not that clear how you can derive in this case and this gradient of the free energy and what it suggested that the solenoidal couplings can cancel out but but yeah but i so we are going to consider the two cases but uh considers in principle just the the case of a complete diagonal so this first assumption means that the solenoidal coatings between elements are or can be approximated as zero okay and the second assumption is the and okay so we have a flow of external states that's pointing in the but we want to interpret internal states as minimizing this free energy so what the free energy small purpose purposes to solve this so the the proposal is that there is a a smooth mapping sigma that connects the most likely internal states with the most likely external states so this m y is equal to sigma x and and the assumption is that there's a and of the of yeah there's a gradient or there's a derivative of this mapping that exists so the mapping is differentiable and that it is invertible and they propose that the sufficient requisite for this assumption too is that the mapping from b to mx is injected okay and so if this mapping exists then we can connect the external states with the internal states and we know that the external states about flow that points in the direction of minimizing the free energy and so the hope is that now internal if this mapping exists are going also to have a flow that's pointing in the direction of the free energy and the last assumption and this is maybe more controversial is that if one have a strict interpretation of the gradient descent on of a free energy and the the assumption will be that the dynamics of the most likely states are approximately equal to the dynamic to the average flow of the external state so so this is the derivative of the average state and the right side will be the average of the derivative to some so so so this is saying something like this no proximity the derivative of the rate is going to be equivalent to the average of the derivative because the flow is just the the time derivative of the of the system uh so and if this assumption will be true then not only the flow of external states point in the direction of the primary but also the dynamics of the system of the uh are minimizing the free energy and and this is interesting because if we had this if this is true then we have some equation like this so the derivative of uh of time y is equal to the gradient respect to n y of this surprise so the dynamics of m y are behaving as if minimizing the free energy so we so this equation is true we know that the dynamics of the behavior of the system is going to be minimizing actively performing a gradient descent on the free energy and if this is true and we have the mapping that assumption two proposed then we can use the chain rule to find a connection between between the derivative of m y and the derivative of n y so the dynamics of the external states and the dynamics of the internal states just with the gradient of this sigma function so this is just a chain rule of the of derivatives and with this step then we can describe the dynamics of internal states as the dynamics of external states and use this gradient mapping to reach this equation that tells you that the dynamics of the most likely internal states is doing a gradient descent on the free energy so this is the end of the of the of the argument of the free energy so so if this is true then the internal states will be behaving as the free energy about external states and then the internal state will be behaving as if effectively performing uh by a budgeting inference okay and and well we derived that from some of the the steps in in these publications in the monograph of a particular free energy for a particular physics and the most recent uh paper based formats about markov blankets information geometries and so on because they or at least it appeared to us that they were using the same symbol for the derivative of external states and the average flow and because so in some equations there is a connection between the any so for the chain rule they use this the symbol and point and so so this is why we interpret this in the in this way however and also from conversation from the authors uh it's proposed that the the free energy principle is not exactly proposing that the behaviors so i need the presentation of the principle will be that the system is not doing any straight creating descent as this assumption three that we label with this star because it's in part or interpretation but that the an alternative to this assumption star 3 will be assumption star star 3 which is that it is not the derivative of the most likely state that follow the grain of the free energy but that but just the flow the average flow so the average flow of f y is just the average of the derivative of y given v the conditional average with the conditional distribution and and the claim will be that if the flow is pointing so if the average derivative is pointing in the right direction of the free energy then you can interpret that the system is behaving as if on average it was performing a gradient descent so so this assumption on star star 3 would be that not that the system is exactly doing a gradient descent on the free energy but that it is doing on average now if we will have like many observations of of the system then we could see that they on average points to a minimization of the free energy but maybe not if we just observe one system individually okay and then this assumption starts start three will be that yeah exactly that now if the conditional average flows follow the direction of uh gradient descendant on the free energy then the behavior can be described on average as if minimizing the greenhouse okay okay so this is the summary of the different conditions and assumptions of the principle in which we will have competing interpretations at some points like this assumption three um okay um okay this is a lot of steps and it is hard to intuitively have an idea of of what goes what yeah how we can expect from any of them so what we did is we try to explore these in a in a very easy system to solve and and and see what we can expect uh in very simple case no so our ideas the best way to really understand this is to have a toy model the the same place we can have and and see how easy or hard is for the different assumptions to behold to be held by the system okay so we substituted the langiban dynamics that we had before by just a linear language so everything is the same and well we have a factor for the noise but this is going to be this one and and we have we have uh now the f function the flow is just substituted by a linear multiplication so we have a matrix j which is an invertible unreal which is multiplying the state of the system minus some constant raw which is just going to be an offset capturing the the average state in a steady state okay and this system is going to be a non-equilibrium system if and the matrix j is is not symmetric okay so in that case we are going to have a non-equilibrium and we are going to have like for example spiral flows and so on and the interesting thing is that this system has a has a steady state if we let the system run for some time it's going to have a steady state that's going to be a gaussian distribution because if everything is linear and the noise is linear eventually the system is going to reach a steady state gaussian distribution and in the limit of time time going to infinity the average of this gaussian is going to be just this row parameter and the covariance matrix of this multiplied gaussian is going to be this sigma star matrix and and this sigma star matrix cannot be solved uh analytically but it's the solution of this equation 29 it's an equation that involves the noise the covariance of the noise gamma and the couplings yeah and if in equilibrium this matrix these equations can be solved analytically but we are interested in more generally in non-equilibrium systems and this this equation is called a continuous line of lyapunov equation and in principle it can be just solved numerically which is not ideal for us because we want to access to have the connections between the parameters and and the matrix but we cannot in general okay so what what did we do um so in order to have a solution of this equation a general solution we assume the following we assume that the couplings of j can be defined as some diagonal term i and this is negative just to ensure the stability of the system so if if if y if i is is positive and large then you will have a diverging system and then we have a non-diagonal couplings c and we assume that these c terms are small so if we have c square or c to the power of three and these are going to be neglectable terms and in that sense we can so we have that we can solve this equation you know in terms of a power series expansion so we can have like a so this will be equivalent to a taylor expansion that you can see in calculus and and we can have an expression of this sigma the covariance matrix of the solution in terms of increasing powers of c so we have here the linear term the quadratic terms and some cubic terms that we are going to ignore okay and to make things more easier we are going to assume that the noise is homogeneous so sigma is just identity matrix times some constant some scanner constant okay so we choose values of so if y so the conditions one and two are almost automatically um true if y is not even larger than c we have a global attractor and also we can define j to have the sensory motor structure that we want and so we go directly to condition three so condition three is the markov blanket condition proposing that the blanket states or the central motor states play a very special role being a boundary between external states and internalistics so that the distributes the conditional distribution is an independent distribution given the blanket y and x are independent okay so in a gaussian distribution and this condition is true if another leaf the hessian matrix which is the inverse of the covariance satisfies that the decision between y and x decision between internal and external states remember that this can be a matrix if the dimensions of y and x are more than one it's equal to zero and okay so we can very easily compute uh what's the form of discretion with these approximations so we have the question how common are markov blankets now from all the systems in the world and how often can we expect if we have just random parameters to find our markov blanket okay so we want this matrix h which is the inverse of the covariance of the steady state distribution and we can express this inverse again as a power series and this is called a neumann series for computing the inverse of a matrix and it has this form we have again a linear term and a quadratic term and some cubic terms that we are going to ignore okay and so we just need to know c to know how what's h and to know if this corner of the matrix is equal to zero okay so so c is going to be like this because we impose these specific sensor motor structures in which external states don't influence internal and internal done influence external and so on and so for the first order approximation we have this equation 36 okay so so you yeah for a third sort of approximation we just keep this term of the equation and we can see which is c plus the transpose just switching rows and columns and we can see that if sum c plus the transpose the term in the corner is going to be equal to zero which is what what we want so for our first order approximation we have a an exact markov blanket and the markov blanket is always going to be true so this is a good news in principle and however if we if we are a little what if we want more precision and we compute a second order so we get this quadratic terms here we find that the corner of the hessian the one that has to be zero has this form so in principle this is not going to be zero or is going to be zero just for various special cases and if we look at the connectivity structure of the of the system so the kind of sensor motor loop that we require in general is this a canonical loop at figure a so in principle for this structure very few systems are going to have a markov blanket because this is not going to be zero exactly almost never and however there is some other configurations that are going to have a mark of language if we have a circular loop like this case b or we have something like this a structure in c then uh the relevant then then the terms appearing in this equation are at least one of them is zero in both cases and this is going to be zero so v and c are going to have a markov blanket in general but not a and there are second order approximation and so under this canonical flow constraint so case a only a few systems are going to display an exact mark of blankets although about one exception are going to be the circular loops or the symmetric loops and however we we fair we we have to say that um in the case of weak couplings uh the markov blanket is still going to be a reason princeton or reasonable approximation because if we have some equation like this and so so so the first term so so even if we have something like this this is a second order equation and all the other elements in the matrix are going to have first order components that are going to be much larger so even if it's even if we do not have an exact markov blanket this is going to be a reasonable reasonable approximation in most cases okay so this is condition one conditional three sorry so let's go to the next the next one is assumption one uh which states that uh the solenoidal couplings between blocks of state have to be zero in order to have a flow that points in the direction of the free energy or we could relax this assumption and have this this larger blocks but also we're requiring some some zeros in here uh okay and so we can rewrite the the previous solution of the of the steady state equation uh to find an exp a connection between q gamma and j and we and doing that will reach right in the previous equation we can reach this equation 40. and again and this is this equation can cannot be solved directly in general because it's another continuous like upon equation but assuming uh that we have j equal c minus y like and the coordinates are small as we have before we can do the same trick and have a new power series that connects q with the values of c okay and has this form which is similar but a little bit different than the case before um okay so but this case looks a little bit more complicated so first we just computed the the first order expansion of q and it looks like this so so this is q for a first order case ignoring this the quadratic terms of c okay and again we see that there is in principle few cases where then yeah where the assumption is held exactly because we are we're having you we want to make zero all these terms here and if we don't constrain the possible parameters this is going to be true only in very very special and we can see that one way to make zero many elements of this matrix is assuming that they are symmetric if c and y s is equal to the transpose of t so so if the connections from sensor to environment are equal to the connections of environment to sensor then this is going to be zero and in that case we obtain this matrix which has a lot of zeros but still there is these terms that we cannot get rid of so if these couplings are not zero then uh we don't have the assumption of the free energy principle and and this is important because well this is well this is this will be the the connection from actuators to the environment which in principle is something we could expect in many cases and this is the uh yeah and this is the connections from sensor to internal states which is also something we expect in most sensory motor or we intuitively expect from a cognitive system with a sensory motor loop and if we want to make these elements of the matrix equal to zero then we need to have something like this loop c in which these elements are symmetric the js the non-diagonal js are symmetric and also we remove the connections from sensor to internal states and actuators to external state which is intuitively uh a bit strange but but i think is in some of the literature of the friner events with this kind of loop has been used and so and it has probably and it has purpose that this could represent for example the the membrane of a cell that separates internal and external states in this kind of symmetric manner although i think there is some well there's some research on cell membranes indicating that asymmetries in them in membranes are still important so this could be still a potential issue for finding systems that that have this this zero this block that you're gonna help you and if we consider the second order approximations things become spookier and we have many many terms and we see that these elements are just going to be zero in very in a very narrow set of of cases which potentially problematic for for uh yeah for asserting that the principle is in general for it would yeah this assumptions will be general for for many systems and this also gives you an idea that we we are starting this very special case with very weak couplings but having a stronger couplings in principle will make things more complicated and not easier and okay so this is assumption one and we only are going to get this block diagram they're very specific circumstances and this as the only exception being this symmetric loop with this very very special and so so this is so imp so in conjunction the markov blanket exploration and the and the exploration of these soldiers accomplish uh will indicate that under the class of systems that we studied the the free energy principle will be will only emerge exactly for a very narrow set of systems okay so so this is um so interesting to think about how general the principle is or how can we expect it to be to be found in in in different classes of systems although we just restricted our analysis to these linear ones um okay so the second assumption and connected to the question of how general not not how general but how informative the system is and the second assumption is is that we have this mapping between internal external states and and if we compute the most likely international states given a blanket for a gaussian distribution we can do it very easily to computing a conditional average and so these are the questions you can find it in any textbook multivariate gaussians and we find that if this requires to compute the generalized inverse of some matrix and and we find that we can in linear systems and in some cases if the kernels of these are correct and we can derive the mapping but these mappings if we have the correct dimensionality will exist very broadly so this mapping is just going to be a linear matrix multiplication of the relation between m y and mx so so this is good news and first of all this mapping exists and it can be expected to be very broad if the dimensions of these variables are are correct although maybe yeah maybe we should know that in some of the literature that some of the verbal descriptions say that this mapping and it's a consequence of the marketplace but what we found here is they are principally independent they they could emerge even if there is not a markov blanket so at least in linear systems uh and also we we discovered that this result was also found in independently but by lance the costine award that is going to be published soon and it's i think that they they got to the exact same result as we did uh okay so so going to the last assumption which is maybe the most complicated and so the last step of the free energy principle was to say that we know that the flow of external state the average flow y is pointing in the direction of the free energy and this means that the system is behaving as if and i described that there is two possible interpretations for this an extrict one and a relaxed one so let's explore first the strict gradient descent interpretation which would mean that the derivative of the average of the external states is equal to the and we can compute that very easily in our linear system this this equation is the derivative of the average state and this equation is just the average flow and what we did for comparing both is just generating a dummy variable this tilde m which behaves as equally as the average flow so this this variable tilde n will be a variable that strictly minimizes the free energy right so so we have so this is the real dynamics of this of these study of these most likely states and this is what these most likely stages will do if they were exactly and under or we copied approximations we can see that we can get to simplified version of these equations and we can see that both of them are a bit different at first the real dynamics has a noise noise term because the fluctuations are going to affect the dynamics because the the observations of the blanket are true by noise so this is going to be a noisy dynamics and this is not and the second difference is that this term changes the sign here so here is positive and here is negative and this seems strange but the the reason of this is because and this so this is the real dynamics which depends on the previous state and average flow it's his story independence in some sense it just depends on the previous state of b but it doesn't depend on the previous so to speak so so in that sense the the structure of the of the flow of this variable of the dynamics of this variable is going to be different and in this case this difference is captured by this constant that has a different sign between one and the other and if we run a simulation of both of both equations we find that the behavior is very different and so this is the behavior of the real average variables so this is a system with two and you have the symmetric forms because the the we restricted the couplings to plus minus 0.1 and if we represent this tilde m so so the system that strictly minimizes the free energy we have this very very different behavior which has this kind of a random walk and and this is because this this variable is integrating the fluctuations of of the other one and so the so the error keeps accumulating and and the behavior is very different so in principle it's unlikely that the and and these results were similar for many combinations of parameters but in principle is unlikely that the dynamics of the most likely states are going to be um so that the average flow is going to be informative about the actual of the most likely states okay so this is the strict version of the assumption the relaxed version would be that only the average flow follows the gradient of the free energy and if we if we interpret this we find that there is two problems at least we thought that there is two problems and the first problem is that a key step of the free energy principle uses the synchronization manifold so the sigma function connected external and translates and then derives a chain rule connecting dynamics of my with the dynamics of internal states and and this leads to saying that the flow of the internal states is following the gradient of the free energy okay and so if we say that the free energy does not truly apply to the time derivatives of the most likely state but just applies to the flow then we cannot use the sigma mapping to the right this equivalence between what the external states are doing and what the internal states are doing and so we cannot so this step becomes problematic and if we want to connect the flows of internal states and the flows of external states as the as the energy principle proposes then one will need a new mapping because and this mapping are not the same and we computed in nonlinear system the two mappings and we find that the the mapping has a different expression and it is different from the mapping that's the gradient of the mapping that's used in the literature so so in principle this relax interpretation will require uh a different mapping that's different than sigma because we are not interested on the flows of the most likely states but on just on these average flows and okay sorry yeah uh and well and more importantly uh another problem arises which is um yeah sorry so so if we have an a street uh sorry one second yeah if we have a strict interpretation of the uh okay sorry i lose this uh yeah if we have a strict interpretation like this like a of okay internal states are following a gradient are doing a on the on the free energy so the derivative the temporal derivative is equal to the gradient of the so this is uh this is a definition of uh like a textbook greek in descent okay and this is fine if we have this equation we know that the system is doing a gradient descent so the question is if we just have this equation here so the average flow is pointing in the direction of the can we really interpret that the system is behaving as if minimizing the free energy so how informative is this average flow about the real behavior so we try to explore this so first and so well first in the simulations we see that it may be little informant there yeah it would be little informative even if we had a different mapping and maybe this average flow has little information about what the actual dynamics is doing and for exploring this further and you can see it in the in an appendix so for for a preparant like this the second appendix uh we try to to explore what is the connection between the average flow and the actual dynamics of the system uh in a very simple system so we restricted our system to a bi-dimensional case with this non-equilibrium behavior this is so this the flows of the system are have this spiral behavior with under noise are going to be rotating in this muscle so we have two variables y and b and so this is the flow the actual flow of the system and we ask okay so in this system what is the the dynamics of the most likely so this derivative of m y and given v and we find that we have this negative slope which is uh well you have this this pink area because this is a random variable which is going to have noise and it's going to be noisy and so on but on average we have this negative slope meaning that the system is going to be minimizing it's going to have an attractor at zero right if you study dynamical systems and have a negative slope of a derivative or at least on average you are going to minimize so if the state is negative you are going to reduce it you are going to increase it sorry and if the state is positive you are going to reduce it so you are going to have an attractor at 0 which is what we see in the actual system but at least for some combinations of parameters which we were like half of the possible combinations of parameters that we tried if you compute the average flow it gives you the opposite behavior here the slope is positive meaning that if you have a system that follows this slope is going to diverge to ma plus infinity or minus infinity so this is intended as an example that the average flow can be a very poor description of the actual behavior of a system if what you have is a stochastic dynamical system and okay so so in general we cannot assume that under all circumstances the marginal flow is going to be informative as the behavior of a system and if even if you can derive all the steps or of the free energy principle it might be the case that in some systems the average flow that's pointing is pointing in the direction of the free but the system is not behaving as minimizing the free energy you know you can see it in this example uh the system is minimizing the value of y square it's going to zero not the here the value of of these variables is going to zero but the average flow points in the opposite direction it tells you that if the average flow if you interpret the average flow as what the system is doing then you will believe that the system is maximizing the square value of y for example because it's diverging and what it's doing is just minimizing it because it's going to zero okay and so in principle if the if the dynamics of the most likely states and the and the average flow are different then uh yeah so they can they they are going to be in general difference and in that case they are not going to be informative what about the behavioral system except in various maybe very specific cases as mean field approximations or this kind of thing and and this difference is because the these marginal flows arise from a decoupling of a previous history of the system and they are not considering the effect of the effects of fluctuations and fluctuations okay and okay so just to finish and maybe let's give place to the discussion and when this work we explore the conditions to and the assumptions that are needed to derive the free energy principle in linear non-equilibrium stochastic systems and and some of this has been explored for specific counterexamples as a recent work by martin bill and others but i think it's interesting that we did this for a general class of systems so it's not one example that we saw numerically but we solve a family of weekly co systems and we see how general or not are the different and so first we explore that we explore how general are these conditions and assumptions of the principle and then we study under what conditions can we expect and the principle to be informative about the behavior of a system and first we found that the markov blanket condition emerged at least as a good approximations in the case of weak weak couplings but if the couplings are not super weak are not very weak then we maybe we cannot expect to be exact in yeah in general and however solenoidal couplings are generally present in all blocks of the system so so and they you are going to be serious in very very special cases as this symmetric loop so maybe a direction of two of the for the theory will be to yeah to explain exactly what happens when these these and coppers are not zero and and what yeah what happens in that case and and so in conclusion we see that only very specific structures are going to satisfy their requirements uh to derive the the to the right that the average flow and the second point about how informative the principle is we saw that the connection between the free energy and the marginal flow of the system with the behave the actual behavior of a system is is in most cases very very weak or even contradictory as one can point in one direction and the other in the other so we found that this in general these marginal flows are not capturing the the actual dynamics of the system and and this is because the this marginal flow is ignoring the history of interactions and we could intuitively think about this so we will have a so we can imagine a system that has many trajectories and if we fix the blanket at some point it just means uh having different trajectories that are going to pass through this point and the average flow is just going to compute the directions or the direction most likely derivative at that point well the average derivative of the system at that point but an any individual system can have a very different behavior which is represented by black line here and so an intuitive example for this uh could be the uh to distinguish between the the population of an individual organism organization and the world and the behavior of a population and so you could think that the on average uh a population of organisms is behaving in a way that's maximizing evolutionary fitness but this can be largely an informative for describing the behavior of individual organisms not that some sometimes the kind of criticisms that we hear and to some proposals in a person evolutionary psychology and this kind of thing right so so even if i at an average macro level you can detect some tendencies in many cases these tendencies can be uninformative about what an individual is doing right and so so we think that it's interesting to to think about uh if and when and in which cases the the the conclusion the conclusions derived by the free energy principle are informative about the behavior of an individual or or groups of individuals and so on still we we have to say that um even if we did this what kind of critical evaluation we are still we find that the the the motivation of the free energy is very it's very suggestive and and we think that the the aim of connecting ideas from variational inference with the dynamics of complex self-organization system is tremendously important as it could allow to apply the machinery of magician and information theory to describe systems that are intractable in practice but but we also think that it's some how difficult to just write a theory and without a connection to specific models right that can allow you to inspect in more detail some assumptions or some consequences of the are difficult to grasp intuitively in many cases right and and so so in principle we find that the yeah that it will be interesting to yeah that exemplifying the steps of the theory in a specific model is very useful to to yeah it's very useful to understand the connections between steps and how likely or not are different steps and that this can this kind of exercise can be only useful too and yeah to to to do yeah it can be really useful to to and understand the kind of uh the difficulties that we face for even a theory and just one so this is the the preprint that we just published how particular is the physics of the free um thanks a lot thank you thanks for this awesome presentation there was a lot there and it was incredibly well presented so for those who are watching live i would invite them to write questions in the chat but first um chris i'll just pass it to you for an initial comments if you would like oh unmute then please continue wait chris you're admitted and then continue yeah so miguel that the we even though we we found these laws in a particular kind of technical presentation we've seen we still find that the kind of the the ambition uh uh free energy principle is laudable right and we're really it's something we really want to embrace but yet we feel that it needs uh some future development right so i mean i summarize there's two issues that miguel has presented here the first one is the problem about trying to characterize the interaction of an agent with its environment so you can do this in two ways and one typical way is to draw kind of connections and direct functional connections between an agent and environment right so the sensors and the motors and so on uh well there is a different way of understanding those interactions in terms of the testing or the covariance matrix which kind of describe correlation system rather than direct connections and i think what uh carl has done is kind of really work from the hessian upwards rather than considering the functional connections that underlie and underlie those two underlie those right so he wants to assume there is a markup blanket and wants to start there and and then he's not so concerned about trying to work out the class of systems that meets those conditions and i think that's a valid thing to do it might be a very special set of systems but i think carl will claim that's the kind of systems i'm interested in i'm not really interested in those other systems right and what we've pointed out is that we can't find those uh very generally in in the class of uh and the second major issue that i think uh miguel's work points out is this decoupling between the description of the deterministic of the system versus the average dynamics of an ensemble of systems right so in prior accounts of the free energy principle you actually write the deterministic dynamics into your description so when you're using for energy principle to do cognitive systems you are describing the certain deterministic dynamics directly in terms of your expectations and and uh and flows on your expectations but what the particular physics approach wants to do is try to try to find that deterministic dynamics within the physics right it wants to see where it where it would lie within the kind of general stochastic dynamical systems theories of physics and that's where the problems lie right so it's very hard to go from the the physics to the description in terms a good description of the system in terms of the kind of uh dynamics on expectations and and so on in that system so we find this problem of a decoupling between the description and the dynamics of the uh the the moments of a system versus the average over an ensemble of systems this is where where the problem lies right so the over ensemble systems the free energy might be a good description but it's not connected to the causal explanation of the dynamics of a specific system so you can't attribute kind of the uh the right kind of mechanistic description to that particular system that you would like to uh and yeah so i'm that's kind of my summary maybe thank you for the succinct summary that's very helpful for many i'm sure claren if you'd like to um unview sorry i heard a feedback when chris was speaking so if you can unmute or uh yep okay yeah introduce yourself and then give any comments you'd like and then i'll go to the questions yes bro so hi i'm ben i'm obviously an author on this and i'm currently now a postdoc in oxford but i worked very closely with uh chris and miguel at sussex for a long time so yeah also simon later had something else scheduling conflicts i just jumped in for the questions but yeah so i would just like i think chris did an excellent job of explaining the issue with the average versus the dynamics of the average but i would just like to go in and kind of reiterate to me an important thing about the markup blankets is really this idea that the markup there is a huge distinction between the sort of statistical independence relationships and this kind of like causal actual relationship you get if you write out the dynamics and so it's quite easy to confuse these two and indeed i think we we naturally conflated these two a lot until we started writing this paper and a lot of the literature also like conflates these two and you know they have these intuitive diagrams of you know like the bacteria with its cell wall and everything and that's the blanket and so forth but that's in this causal level of the description like in the things cause we have to pass through the blank but that's not the same at all as the statistical description which is you know the statistical independencies between you know the blanket and the intel and the external states and i think yeah it needs quite a lot of thinking about what extent these causal mappings actually hold up and sort of actually sustained by the statistical mappings or vice versa and we've found in linear systems there's a lot of cases where they don't and they don't seem to matter at all and so yeah it takes a lot of kind of different intuitions really i think need to be built about what these markup blankets mean and actuals will cause a level thanks for this comment and again people can write questions if i could just actually add something about the difference between the causal structure and the correlative structure you could imagine on a market there's companies that are causally related to each other but their stock prices are not correlated and similarly in the cell we often make networks that are based upon like gene co-expression so genes with correlated expression and then there's gene regulatory networks or protein protein interaction networks that are like proteins that touch but those proteins usually don't have correlated expression because you might up regulate one that then represses another so they're not expressed in a correlated fashion but they have causal so and it's people slipping between oh gene regulatory network you know co-expression interactions certainly conflating the causal structure of a system with the physical structure or the correlative structure is a broad category error and i think your work does an incredible job at highlighting the difference and where slipping between these easily confusible modes of thinking about networks and then also conflating that with reality or all of reality just very well done to these points yeah thanks yeah another example would be like the difference between structural connectivity and functional connectivity and effective connectivity in in in the brain right you can have some structure but and then you can have statistical correlations but even then the dynamics can behave differently because you have like these asymmetries in in how one variable is connected to another and on the opposite end and this is also not work the relation with the solenoidal flows that capture something different than so yeah so it's you know very complicated to disentangle all these different levels and but you just justice to the office i mean i think they're aware of this difference obviously uh carl's group is very aware of this difference but they they feel that the theory sits above that right at the correlative structures and not considering uh those causal structures i think maybe some confusion might have emerged in previous papers but in recent papers i think it's very clear that they want to sit above the causal corrections cool so i hope that by bringing the math to examples of like the market evolution um and fmri we can hopefully have a lot of people giving their perspective so i'm gonna just go with a question from the chat and then i have some questions that i wrote down so uh first question thanks a lot i was wondering whether you would assume that going to non-linear systems the limitations that you find would be even stronger or would there may be an argument that there could be non-linear systems that fulfill these assumptions so what will relaxing the linear assumption do for the math that you described today okay well that's that's a difficult to to answer a question in general i think so so okay so so first is is i think it's it's difficult to well it's difficult to solve because uh in nonlinear systems in general we cannot find the kind of analytical solutions that we have here right and and even in the case of linear systems we found that we can only solve the case of weak copies and in the case of weak couplings we could do the trick of considering different orders of expansions and we saw that including stronger couplings make things more difficult rather than rather than easier okay and intuitively uh at least my intuition about non-linear systems is is that the case may be my may be similar in many systems right right having strong that yeah having more intricate couplings more intricate dynamical couplings could add extra turns uh to some of the matrices that we are expecting to be zero like in the markov blanket and but still this is not something we can say because or you will have i know unexpected effects from chaos for example and i think that's something that carl proposes sometimes and but i don't know i think maybe one interesting direction of research will be doing a similar expansion to the one that we to say okay we have an almost linear system but we consider very small effects of non-linearity and what kind of terms this adds to the to the matrices that we are so this this could involve a very different analytical solution but i i think there is some up for coming work and from uh yeah from some people in cars working in that sense so so i think it's an open questions but but i intuitively think that in many cases this is going to be challenging to to continue yeah to see that non-linearity is going to be a solution um i mean you feel the intuition is that you might be able to solve it in specific systems uh but you know you start but the idea that you'll solve it in a classic non-linear system seems seems uh doubtful right but you might be able to find examples which you augment with nonlinearities that meet the criteria maybe if you constrain yourself to a particular evolved uh living system that might be okay right but if you want to say something more general about the emergency self organization in the classroom physical systems they perhaps run into trouble two thoughts yeah we'll go ahead and first yeah i agree with chris and that obviously with non-linear systems you have a much bigger space of systems and so you can probably find you know edge cases where all the conditions match perfectly but i mean the general intuition you dynamical systems is the non-linear systems are a lot harder to deal with in general and so i'm not going to probably you know if it's not satisfied in linear systems there's no obvious reason why it should be satisfied in non-linear systems and especially the issue with the the dynamics of the average versus the average of the dynamics like the case we did where you have a linear gaussian system is probably the ideal possible case for this because obviously the average is the linear operator and the for the gaussian case the mode and the actual average are the same and so you have all these nice properties none of which will hold in the nonlinear general case and so i think yeah in for that non-linear system is almost certainly going to be much much worse for that specific you know issue yeah also to to on that i think so so maybe so one question is is whether there is linear systems in which some of the problems are going to go away but maybe the the right question is what happens in the kind of non-linear systems that that we expect in living system right and what we know from living systems often involves bifurcations or critical points that that lead to a long range correlations and correlations that that propagate through the whole system and this kind of thing that can be very difficult so so the kind of isolation between internal and strong states can be i think can be challenging in in when this is the case thanks for all these excellent points um in complexity we often talk about non-linear as being like non-elephant animals it's like all of them so it's even harder certainly to make a general stance and baron that was a really interesting point that as the state space of what's possible increases there may be more and more edge cases and that kind of brings it back to the question about evolved systems not every parameter combination of cells is going to be functional and so maybe we're only dealing with that vanishingly small amount or some other i'm not going to you know go into technical details on the fly but that could be one aspect and also it was such an interesting point during the discussion when you explicitly brought it back to evolution like fitness is the law of the land in a way at a level that doesn't explain motor behavior in the moment to moment their nested time scales and so of course there's motor behavior that's maladaptive and long run you wouldn't see that under stationary ecology and dot dot dot dot dot you start adding some assumptions in and it's just i think a very fruitful mapping that will be explored for time to come with the relationship between popular population level claims under stationary environments population level claims under dynamical environments and then individual level claims under stationary and and changing environments so it's just a lot of topics that come into play and to have them integrate these kinds of um points that you're raising which are often implicit errors in other areas they're being drawn together and brought to the forefront in a way that is quite novel okay again since the technical details i believe stand best in the presentation as you provided as well as in the paper we can take any questions but i think that will be the place where the discussion continues i just kind of wanted to ask a few other questions to provide a little context so for maybe whoever wants to answer how did you come to be studying the question this way were you working with the fep and exploring new kinds of math or did you have more of experience with the mathematics and then see the fep coming up onto the radar like how did um each of you come to be working on it this way okay yeah i'll go because i think i'll go over so i mean i i wrote a review a few years ago now of the original energy principle at that time it was it was quite an enigmatic uh principle not many people understood it i mean the reason i originally i didn't like it i didn't like the idea of the notion of representation seems to be pregnant in the idea of the free energy principle kind of representing uh generative models but i actually came to like it's a cognitive framework and i still do like it as a as a cognitive framework uh and i still still think within the regime of describing essentially motor loops abstract sensing and cognitive systems it's very very useful but then obviously then there's been a rapid development of the free energy principle into uh into into a grander claim right to find this principle in the in physics in a class of physical systems uh and that's where kind of i started working with miguel and gail's got a real strong background in statistical physics and that's basically what's led us to this to kind of really go back to some of those so i do separate the two things i think there's the still i'm very fine the the cognitive account very appealing even though we found these problems in the in in these physics uh examples yeah yeah i all the uh well i for like yeah for some i think medium i started doing involved with the the principle more recently i yeah for sometimes that is something that i had on the radar but but actually understand it what it meant like it's always seen seemed kind scary and challenging you know like the equations and and i guess that part of the motivation for doing or doing this work is is it was to as a way of understanding what what the theory consisted of because you could go through the mathematical descriptions and it's often really hard to know and what what they imply or what they mean exactly in the in the sense that they are complicated maths right and i guess that the the during my research career that the way i have i have a hat for for understanding complicated masses by building models and this understanding by building philosophy that i so so i think that the initial attempts to to build this system it was an attempt to to really understand okay this step i can for us mathematically more or less what it means but but let's have a model and change parameters and play with it and and so on right um so so that was more or less the the journey that and and once having that first having the model then say okay can i have something more general than just a some specific parameter so let's try let's see how we can have an analytical solution i mean and so on so so yeah so i think yeah exercise is very fruitful both on a personal level and also on a scientific level because you can on your way to learning you can develop things that can be useful for the theory more in general awesome thank you for those answers and also and i was just getting a feedback right now from you miguel so just mute while baron gives a yeah okay go ahead yeah i just want to follow up on the girl's point really about just the importance of actually trying to build these models because i think from my perspective a lot of developments and problems we discovered basically came from actually looking at specific models and thinking hey this this doesn't quite work out as it says it should you know hey because basically things have just become an awful lot more clearer if you actually try and build them rather than just staring at these sort of symbols on the page and say i think that's really the key to it to a lot of it you're simply trying to construct a system where this works and then say hey you know if things aren't going quite as you know the particular physics monograph says they should be cool and it's so interesting and maybe even appropriate that action became relevant for this inference so it's not just conceptual because we must be engaging in the literature and conversation in the work and going through it ourselves in order to infer deeper we can't just wait at square one and look all the way to square one thousand we have to be working along the way individually and maybe as a group too so here's a question from the chat brilliant talk wonder if carl and others will respond to challenges presented in this paper also because the paper is written in such coherent style wonder what would you recommend for beginners to study in the fep nice question what do you wish you knew at the outset even if it's a new resource what courses or topics would you recommend having a familiarity with or what are the on-ramps for somebody who's not in area of research and practice how can they best be on ramping themselves so i recommend baron's github he's got a nice repository of papers uh which go from uh very introductory papers to very sophisticated papers in nicely organized ways i don't know if you want to provide that is it public publicly accessible oh yeah publicly yeah i could post that in the chat or something if you want yeah say hello in the youtube chat and then i'll make you a moderator and you can post links so yeah yeah so then continue on that yeah i mean regarding the actual question about what to sort of start with i think that depends a lot on what you want to do so there's kind of three big separate topics in my mind is that sort of the fep there's a hardcore maths of the theory as is presented in particular physics and for that i'm not sure there is necessarily a good beginner's level resource at this point like there is essentially i mean our paper tries to go through quite straightforwardly and then they have you have the thomas par markup blankets and stochastic thermodynamics paper which sort of covers the main points but it's not like a tutorial kendrick states like this is xyz and then really it's just the particular physics monograph is where most of that information is like there isn't yet a really nice laying out of it and obviously we kind of try and do that in our paper but obviously we're also focused on the actual points you want to make it's not just like specifically tutorial yeah i mean the other two things obviously the active influence sort of discrete database active inference that people talk a lot about i think the best resource for that at the moment is probably landslide the caster's review on that and then there's yeah just then essentially there's some several of carl's papers which i think i talked about in my github repository which is going into more detail but lance's review for that is really good and then there's a sort of more continuous state space predictive coding style which i think i still i think chris's his 2017 review is is the best for that to be honest so yeah yeah i think chris is one is the one if you wanted to understand the whole the theory and all the math of the whole walks through very nicely where profiles like if you want to build a predictive coding network which works and you have that is the way to go if i can ask one note before miguel's answer when you said newer science do you see them as just historically juxtaposed or do you see some of these approaches as encompassing one another or being prerequisite to one another so i think the idea is that the sort of fep in the particular physics monograph encompasses all of them in that there you know the idea is that these are just specific models you know you have a spec have a specific variation identity specific generative model and then you work through the math and you get these these protest theories as carl calls them out i mean i think in practice though they stand they can stand alone away from whatever the particular physics monograph says because they obviously make you know empirical predictions about the brain or whatever and so you can actually test these and so even if all all particular physics is wrong then these things could be right and vice versa if all particular physics is perfectly right and these things could just not be what the brain is doing it could be doing something else so yeah i think they stand alone although in theory the particular physics monograph is kind of meant to subsume them all thanks for that response and miguel to the original question what would you recommend a beginner to be learning and looking into uh yeah i'm not so sorry for me so i think yeah i started with thomas bar paper about markov blankets and promising geometry and then going back to the to the monograph of a particular physics and and but but it was it was challenging because yeah it's like there's the maths are complex and and yeah it was and i kind of liked it was useful to uh martin bill's paper a technical critical of different uh yeah of the free energy principle in the sense it was very technical and formal but it was useful that they tried to encapsulate the different conditions in a way similar to what we do here and i think that was helpful to try to to differentiate different the steps which are not always so clear in in the literature so so yeah i think yeah i think there is at least for the part regarding the free energy principle as described in the in in a particular physics it's and yeah we need maybe better tutorials better yeah more materials that are focused to us introductions here thanks for these responses and i know in our acting lab and in many other associated efforts that the focus on accessibility and communicating these ideas in a way that's going gonna be helpful to people from many backgrounds is a top priority so way back in the introduction you know 120 time steps ago you talked about new visualizations and it's definitely the case that paper after paper will use some classical figures um let's just say and i'm excited by visualizations and so was curious just maybe whoever wants to go in for it what would the new representations do or be communicate what style or what made you just wanting to explore visually and how is that in feedback with the research okay yeah i think that's interesting i like at some point i keep looking through the literature and it's paper after paper the same the same graph about the brain sensor motor logo and so on and yes sir so at some point it stops being useful right there because the like some like its visualization can give unique intuitions to some extent right so and at some point i uh yeah i wanted to know more about so so it was it's not always straightforward okay there's an arrow but what does it mean or why is this bi-directional or and and yeah part so so the the figures that we have where we developed them with the idea of okay how can we intuitively and ex express these very complex uh assumptions and ask how do you have a figure of a solenoidal flow all um this is really challenging and i think i i i don't i'm not sure if we did a good job there but i think it's in no that i think many intuitions about the free energy principle are are really hard to capture because they are quite complex and i think it's interesting to try to put in the directions of of having new images that try to at least try to capture that that intuitions and and what do they mean or how to yeah yeah because often i think you have to read the theory based on your intuition or covering with intuitions the part that you don't understand yet especially if you're learning and this kind of thing i think is incredibly useful in general in science right chris or baron would you like to add on that point uh no i mean i just that we found it hard right i mean because there is some these so many different types of interactions we want to represent i mean typically monolithic arrows between agent and environment seems to dominate everything right but when you start drilling down into this you have the difference between causal and functional obviously people like pearl and stuff i've done this in the philosophy right they've been very clear about different types of interaction but i think in uh especially modeling work it's it's become a bit uh clouded in lots of locations especially in the area of the free but yeah i don't know whether uh i think miguel did a really good job but it does look complicated the the final product is complicated looking right but it does do justice uh to these different types of interaction yeah baron anything on that or no yeah go for it yeah no i i would just say that i agree like i'm not naturally a visual person who actually looks at the figures i think just what miguel did was actually really good and it's something that if you actually look at them and try and understand them it will yield a lot of dividends in terms of quickly getting to grasp on precisely what's being said what's not being said because yeah it's very easy to have into this sort of hazy interaction without knowing what that really means it's the same again as a geneticist it makes me think of oh you know nodes are genes and edges are interactions which kind of those multiple classes and does one paper use it one way and go to the methods and it's one way in this paper and it's another one in another way but it's being cited and so it takes a lot of subtlety to pull back to that question about what kind of interactions are being discussed and then chris what you just said there's like um it's a high praise for the work that it does justice because that's what we hope the bleeding edge does for the rest of the knife and the handle is represent visually on a manifold that doesn't deceive it's not drawing hard lines where none exists it's not making false positive or false negatives or false claims visually because visual rhetoric is how many people will learn and think about these things and so that's all it makes all the difference whether there's a bi-directional arrow graphically or not because people are going to unpack it especially if they're not going to go into the nth level of math they're going to unpack it literally and with the knowledge that that's what the expert represented in their um aim so yeah i really do think it's particularly important for this field because there is a separation between this technical morass that sits at the bottom and obviously the the ambition and what it promises so so many people are attracted to this particular theory and then but not not everyone's got time to go in and decipher that so that there's looking at the level of these figures and and trying to make you know discourse and so on at that level so vital that those figures become very clear and very pointed in what they tried to communicate and any kind of lack of clarity is just going to send the whole community off in the wrong direction particularly if you you're not the kind of person who wants to go in and do the statistical physics more of the philosophy or or cognitive science of these systems so it's vital that we get them right in this particular field particularly i think yes it's uh i think there's an xkcd cartoon you know about where the fields are arranged but especially when working with the theory itself it should be lean and as low dimensional as possible and not anymore because otherwise it will be unpacked in ways that are just going to be completely discordant but if it can be represented in sort of like a build in or like an opt-in at this level instead of trying to explain backwards from what we could do in the future in the math we need to start with the axioms and clarify what they are and then unpack those and then it's like just like with any other area of theory did you disagree with the axioms or did you disagree with how they were applied because those are the areas where there's sort of controlled space for discussion but then something like i don't think it would work that way in a human computer interface it's it's neither here nor there for the level of rigor that you approach the questions with today um so i can ask any other questions or we can see if anyone else posts in the chat but that was actually one of my questions was are those all the axioms how do we represent axioms or how do we even know what those axioms are without getting into like too much of a whole you know incompleteness thing what are the axioms were they what you put in a paper and nothing else those are necessary insufficient or is there some other stuff happening actually it's our ambition just to say before i think i'll let michelle answer that but the the we have an ambition to write a kind of more accessible version of this paper which hopefully we done in the next few months so we want to do exactly that we want to draw away from the heart statistical physics and have this paper underneath it but then even be more even more clear about the kind of overall narrative of what we're suggesting here uh and we'll we can answer those questions that you asked very directly hopefully but for the specific case i'll let miguel uh answer up on you and then continue thank you yeah yeah yeah i think that the the pre-energy principle is um no it has been an evolving theory and there is different versions of eating through the different papers in the 10 years that it has been been proposed propulsion and at least for me it was uh uh a difficult job to to try to get to to someone to summarize the theory in that way or to have the exact connections of steps because also sometimes the some descriptions are verbal but uh and yeah i think that's yeah that's uh important yeah an important avenue for continuing the research in this field is try to to do more accessible descriptions or more or try to encapsulate them in in this way or try to yeah to solve this sequence more clearly or at least for an introductory purpose right yeah it's been it's been very much a very uh a living theory the french people i found this when i wrote the review back in the day which is that there is this development that's very clear in the uh in the series of publications that comes out and i think things things have changed obviously as the as they develop so trying to see the consistency across those types of things um which is the fundamental action which is later introduced and so on is very difficult actually and again i don't think it's a i don't i don't criticize the office of this because i think it is a living theory it's something that's kind of really evolved in the literature it's very complex one so it had to be like that but yeah hopefully as we go forward in the next few years we can become more clear about uh these things and we'll be more and more people doing reviews and more critical evaluations like miguel's done which will will you know further ground uh this theory in a place where we can all agree on and agree on the axioms baron and then miguel okay yeah i just want to say that yeah we should definitely not consider all the axioms and the free energy principle at the moment to be completely sort of fixed in stone like i feel this is still living and still evolving in many ways and there's you know some cases where there's questions about do we actually need this axiom can it be more general than this can it be less general than this and stuff so i think there's going to be a lot of more foundational discussion around the actual axioms underlying it so and you know now the people are just really just starting to tackle in a particular physics in a really deep way that i think that there will over the next few years be lots of discussion about this and hopefully in a couple of years we'll have a much more precise statement of you know what the fop says what is needed what isn't needed and you know how everything fits together but at the moment it's still in a fair bit of flux like because really deep critique outside of karl's group on particular physics is kind of just really starting in the last year or so thanks for that miguel yeah yes i also always want to add that okay so so we focus on on the free energy principle and and this claim that dynamical systems or self-progressing systems behave as minimizing the primary but i think that's only yeah one part of the universe of clearance investment you have the active inference you have the design of systems that explicitly minimize the free energy and that's a completely different field and and often this is also mixed and not sure and yeah and there's sometimes a mix so which is the the actions that i invented in specifically papers no but where are you where you are sitting in your work is not always clear for for anyone because of this this evolution of the theory and these interconnections and so on and but i i think that yeah like uh becoming more mature as they feel becomes more mature this will be clear because sometimes this leads to confusion about what the theory means no uh when people say okay the free energy principle shows that the every living system is minimizing print oh okay but um it's yeah so so it's different when you think of agents that do do it by by design or when you try to extend the claim to other systems so there is these different points of view and these different sides of the theory that that yeah i i hope yeah i imagine that they are going to evolve into more comparative corpora compartmentalize or even if they are interconnected but in a way that they earthly area the borders of the different planes and so it's going to be uh an fep metaverse that is appropriate for the online age honestly because when i think about how the needle was pushed here github and interactive computational notebooks pre-prints like yours live streams like the one that we're on a discord server you know a reading group that was arranged by maxwell and others 10 years ago 20 years ago it would have been like what and it certainly wouldn't have been accessible globally so to see how the material and the informational culture and also social culture is changing in a time when participatory science and open science and all these kinds of transdisciplinary ideas are coming to the fore it's just an awesome contribution so um i'll offer any final thoughts from the chat and each of you a final um word but just to say from the lab that we really appreciate your willingness to you know jump on on short notice and work on it and share it in this way and also with these exciting directions that you've planned out and i'm actually going to say i completely agree with you and i have to you know shout out to people like maxwell ramstad who organized these discord servers which are you know really promulgating this debate uh and why communities which are really valuable uh as infrastructure for us people like us to come in and do this type of research so yeah i completely agree with that this this new infrastructure there is really important for this type of stuff aaron any last comments and then miguel of course yeah final word you know i just want to echo what chris said really about just the sheer importance of the reading group max were organized because i think that's where you know i i learned an awful lot about how particular physics works and you know how everything goes through in that monograph and i think a lot of people did there as well like i think that reading would have really built up enough expertise to you know make papers like this and obviously similar to modern stuff and just generally get everything together probably lots of stuff in the next year or so coming out actually made all of that possible really and so yeah also this is a family like if you're interested in learning more about the fop like email maxwell because he has recordings of all the reading group meetings and so that's probably going to be a really good for a good another lockdown i think yeah so yeah a lot of records will be lots of material there yeah if you want to really drill down into technicalities and those recordings will probably be really good as well great and um miguel if you would like to give any last comments um yeah yes well yes yes as you said as i think yeah um partly as chris and bryan say that i i cannot join late to the group and just attend to the last sessions but it was really helpful to to dig through the some of the discussions that took place there and i think in general what so i think in general yeah like what could make the field advance is to to make things successful in different ways and and this can be through resources through communities and also through more transparent models or more easier to follow guides through the maps and i think the part of the difficulties in the field is that there is this a disconnection between people that is doing mathematical work and people that is doing philosophical work and not only here but i think in cognitive science in general or or even more deeper abuse of systems biology and so right and that this multi level yeah i think it's very valuable this this efforts in in in trying to make things more accessible and easier to follow from yeah for people for from multiple disciplines i think that's a very valuable effort that we should try to do collectively because that's the only way and thanks a lot for hosting the session we always talk about tools ideas and people they're a triple helix and so thanks for making it happen with all those today thank you very much for that peace